

Form factors of XXX model

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1 The XXZ Heisenberg spin- $\frac{1}{2}$ inhomogeneous finite chain

The Hamiltonian is given by

$$H_{XXZ} = J \sum_{m=1} (\sigma_m^x \sigma_{m+1}^x + \sigma_m^y \sigma_{m+1}^y + \Delta(\sigma_m^z \sigma_{m+1}^z - 1)) \quad (1)$$

And $\Delta = 1$ corresponds to XXX chain.

The quantum R-matrix is given by

$$R(\lambda, \mu) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b(\lambda, \mu) & c(\lambda, \mu) & 0 \\ 0 & c(\lambda, \mu) & b(\lambda, \mu) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2)$$

where

$$b(\lambda, \mu) = \frac{\phi(\lambda - \mu)}{\phi(\lambda - \mu + \eta)} \quad (3)$$

$$c(\lambda, \mu) = \frac{\phi(\eta)}{\phi(\lambda - \mu + \eta)} \quad (4)$$

and

$$\Delta = \frac{1}{2}(q + q^{-1}) \quad q = e^\eta$$

with ϕ defined as

$$\phi(\lambda) = \begin{cases} \lambda & \text{XXX case} \\ \sinh(\lambda) & \text{XXZ case} \end{cases} \quad (5)$$

R-matrix satisfy Yang-Baxter equation

$$R_{12}(\lambda_1, \lambda_2)R_{13}(\lambda_1, \lambda_3)R_{23}(\lambda_2, \lambda_3) = R_{23}(\lambda_2, \lambda_3)R_{13}(\lambda_1, \lambda_3)R_{12}(\lambda_1, \lambda_2) \quad (6)$$

unitary condition

$$R_{12}(\lambda_1, \lambda_2)R_{21}(\lambda_2, \lambda_1) = 1 \quad (7)$$

and crossing symmetry relation

$$(\gamma \otimes 1)R_{12}(\lambda_1^s, \lambda_2)(\gamma \otimes 1) = R_{21}^{t_1}(\lambda_2, \lambda_1)\rho(\lambda_1, \lambda_2) \quad (8)$$

where

$$\gamma^2 = 1 \quad \gamma^t = \pm \gamma \quad (9)$$

and For the rational case

$$\lambda_1^s = \lambda_1 - \eta, \quad \gamma = \sigma^y, \quad \rho(\lambda_1, \lambda_2) = \frac{\lambda_1 - \lambda_2 - \eta}{\lambda_1 - \lambda_2} \quad (10)$$

and for the trigonometric case

$$\lambda_1^s = \lambda_1 - \eta + i\pi, \quad \gamma = \sigma^x, \quad \rho(\lambda_1, \lambda_2) = \frac{\sinh(\lambda_1 - \lambda_2 - \eta)}{\sinh(\lambda_1 - \lambda_2)} \quad (11)$$

We define the quantum L-operator acting on $C^2 \otimes \mathbb{H}_n$

$$L_n(\lambda, \xi_n) = R^{0n}(\lambda, \xi_n) \quad (12)$$

The quantum monodromy matrix is given by

$$T_0(\lambda) = T_{0,1,\dots,N}(\lambda; \xi_1, \dots, \xi_N) = R_{0N}(\lambda, \xi_N) \dots R_{01}(\lambda, \xi_1) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix} \quad (13)$$

where $A(\lambda)$ acting on $\mathbb{H} = \otimes_{n=1}^N \mathbb{H}_n$.

$$R_{12}(\lambda, \mu)T_1(\lambda)T_2(\mu) = T_2(\mu)T_1(\lambda)R_{12}(\lambda, \mu) \quad (14)$$

with $T_1(\lambda) = T(\lambda) \otimes Id$ and $T_2(\mu) = Id \otimes T(\mu)$.

Moreover, the monodromy matrix satisfies the crossing symmetry relation

$$T_{0,1,\dots,N}(\lambda; \xi_1, \dots, \xi_N)\gamma_0 T_{0,1,\dots,N}^{t_0}(\lambda^s; \xi_1, \dots, \xi_N)\gamma_0 = \rho(\lambda; \xi_1, \dots, \xi_N)1 \quad (15)$$

where $\rho(\lambda; \xi_1 \dots \xi_N) = \prod_{i=1}^N \rho(\lambda, \xi_i)$ and γ_0 is matrix γ .

The transfer matrix $\mathcal{T}(\lambda)$ is defined as the trace of the total monodromy matrix

$$\mathcal{T}(\lambda) = A(\lambda) + D(\lambda) \quad (16)$$

Suppose a reference state $|0\rangle$ called psedo-vacuum

$$\begin{aligned} A(\lambda) |0\rangle &= a(\lambda) |0\rangle \\ D(\lambda) |0\rangle &= d(\lambda) |0\rangle \\ C(\lambda) |0\rangle &\neq 0 \\ B(\lambda) |0\rangle &\neq 0 \end{aligned} \tag{17}$$

where $a(\lambda) = 1, d(\lambda) = \prod_{i=1}^N b(\lambda, \xi_i)$ For XXX or XXZ model $|0\rangle$ is the state all spins up. We would like to find the common eigenstates of the transfer matrices for different spectral parameter λ .

The Bethe equations

$$r(\lambda_k) \prod_{j=1, j \neq k}^n \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq k \leq n \tag{18}$$

with

$$r(\lambda) = \frac{a(\lambda)}{d(\lambda)} \tag{19}$$

The common eigenstates is obtained by

$$\prod_{j=1}^n B(\lambda_j) |0\rangle \tag{20}$$

The corresponding eigenvalue is

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \tag{21}$$

We give the R-matrix $R_{1\dots n}^\sigma(\xi_1, \dots, \xi_n)$ constructed as ordered product of elementary R-matrices $R_{ij}(\xi_i, \xi_j)$, such that

$$R_{1\dots n}^\sigma(\xi_1, \dots, \xi_n) T_{0,1\dots n}(\lambda; \xi_1, \dots, \xi_n) = T_{0,\sigma(1),\dots,\sigma(n)}(\lambda; \xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}) R_{1,\dots,n}^\sigma(\xi_1, \dots, \xi_n) \tag{22}$$

For any element σ of S_n , factorizing F-matrix satisfies

$$F_{\sigma(1),\dots,\sigma(n)}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}) R_{1,\dots,n}^\sigma(\xi_1, \dots, \xi_n) = F_{1,\dots,n}(\xi_1, \dots, \xi_n) \tag{23}$$

Factorizing F-matrices can be given as an ordered product of partial F-matrices

$$F_{1\dots n} = F_{2\dots n} F_{1,2\dots n} = \dots = F_{n-1n} F_{n-2,n-1n} \dots F_{1,2\dots n} \tag{24}$$

The partial F-matrices $F_{i,i+1\dots n} = F_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n)$ are given in terms of partial R-matrices

$$F_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) = e_i^{(11)} + e_i^{(22)} R_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) \tag{25}$$

where the partial R-matrices acting in $V_i \otimes \dots \otimes V_n$

$$R_{i,i+1\dots n}(\xi_i; \xi_{i+1}, \dots, \xi_n) = R_{in}(\xi_i, \xi_n) \dots R_{i,i+1}(\xi_i, \xi_{i+1}) \tag{26}$$

The partial F-matrix $F_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N)$ has a useful expression

$$F_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) = \begin{pmatrix} 1 & 0 \\ C_{1\dots N}(\lambda; \xi_1, \xi_2, \dots, \xi_N) & D_{1\dots N}(\lambda; \xi_1, \xi_2, \dots, \xi_N) \end{pmatrix} \quad (27)$$

Moreover, the relations above lead to

$$F_{1\dots N}(\xi_1, \dots, \xi_N) T_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) F_{1\dots N}^{-1}(\xi_1, \dots, \xi_N) = F_{\sigma(1)\dots\sigma(N)}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) T_{0,\sigma(1),\dots,\sigma(N)}(\lambda; \xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) F_{\sigma(1),\dots,\sigma(N)}^{-1}(\xi_{\sigma(1)}, \dots, \xi_{\sigma(N)}) \quad (28)$$

which means that the monodromy matrix $\tilde{T}$ in the F-basis

$$\tilde{T}_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) = F_{1\dots N}(\xi_1, \dots, \xi_N) T_{0,1\dots N}(\lambda; \xi_1, \dots, \xi_N) F_{1\dots N}^{-1}(\xi_1, \dots, \xi_N) \quad (29)$$

is totally symmetric.

The element of monodromy matrix is simple in the F-basis

$$\tilde{D}_{1\dots N}(\lambda; \xi_1, \dots, \xi_N) = F_{1\dots N}(\xi_1, \dots, \xi_N) D_{1\dots N}(\lambda; \xi_1, \dots, \xi_N) F_{1\dots N}^{-1}(\xi_1, \dots, \xi_N) = \otimes_{i=1}^N \begin{pmatrix} b(\lambda, \xi_i) & 0 \\ 0 & 1 \end{pmatrix} \quad (30)$$

$$\tilde{B}_{1\dots N}(\lambda) = \sum_{i=1}^N \sigma_i^- c(\lambda, \xi_i) \otimes_{j \neq i} \begin{pmatrix} b(\lambda, \xi_j) & 0 \\ 0 & b^{-1}(\xi_j, \xi_i) \end{pmatrix} \quad (31)$$

$$\tilde{C}_{1\dots N}(\lambda) = \sum_{i=1}^N \sigma_i^+ c(\lambda, \xi_i) \otimes_{j \neq i} \begin{pmatrix} b(\lambda, \xi_j) b^{-1}(\xi_i, \xi_j) & 0 \\ 0 & 1 \end{pmatrix} \quad (32)$$

and $\tilde{A}$ can be obtained from quantum determinant relations (15).

It can be shown that the pseudo-vacuum state is left invariant under such a basis change i.e. $F|0\rangle = |0\rangle$. Thus for scalar products of the quantum states

$$\langle 0 | C(\lambda_1) \dots C(\lambda_n) B(\lambda_{n+1}) \dots B(\lambda_{2n}) | 0 \rangle = \langle 0 | \tilde{C}(\lambda_1) \tilde{C}(\lambda_2) \dots \tilde{B}(\lambda_{n+1}) \dots \tilde{B}(\lambda_{2n}) | 0 \rangle \quad (33)$$

We have the expression for F-matrices $F_{1\dots n}$ in the F-basis

$$F_{1\dots n} = \tilde{F}_{1,2\dots n} F_{2\dots n} = \dots = \tilde{F}_{1,2\dots n} \tilde{F}_{2,3\dots n} \dots F_{n-1n} \quad (34)$$

where

$$\begin{aligned} \tilde{F}_{1,2\dots n}(\xi_1; \xi_2, \dots, \xi_n) &= F_{2\dots n}(\xi_2, \dots, \xi_n) F_{1,2\dots n}(\xi_1; \xi_2, \dots, \xi_n) F_{2\dots n}^{-1}(\xi_2, \dots, \xi_n) \\ &= \begin{pmatrix} 1 & 0 \\ \tilde{C}_{2\dots n}(\xi_1; \xi_2, \dots, \xi_n) & \tilde{D}_{2\dots n}(\xi_1; \xi_2, \dots, \xi_n) \end{pmatrix} \end{aligned} \quad (35)$$

2 Scalar products of Bethe states and the Gaudin formula

We calculate

$$S_n(\{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (36)$$

$\{\lambda_k\}$ is a solution of Bethe equations, then

$$(A(\mu) + D(\mu)) \prod_{k=1}^n B(\lambda_k) |0\rangle = \tau(\mu, \{\lambda_k\}) \prod_{k=1}^n B(\lambda_k) |0\rangle \quad (37)$$

with the eigenvalue

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \quad (38)$$

Theorem 2.1. *Let $\{\lambda_1, \dots, \lambda_n\}$ be a solution of Bethe equations*

$$\frac{d(\lambda_j)}{a(\lambda_j)} \prod_{k \neq j} \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq j \leq n$$

and $\{\mu_1, \dots, \mu_n\}$ be an arbitrary set of parameters. Then the scalar product (36) can be represented as a ratio of two determinants

$$S_n(\{\mu_j\}, \{\lambda_k\}) = S_n(\{\lambda_k\}, \{\mu_j\}) = \frac{\det \mathbf{T}(\{\mu_j\}, \{\lambda_k\})}{\det \mathbf{V}(\{\mu_j\}, \{\lambda_k\})} \quad (39)$$

where T and V are $n \times n$ matrices defined as

$$T_{ab} = \frac{\partial}{\partial \lambda_a} \tau(\mu_b, \{\lambda_k\}), \quad V_{ab} = \frac{1}{\phi(\mu_b - \lambda_a)}, \quad 1 \leq a, b \leq n \quad (40)$$

Proof: See primary paper.

3 Quantum inverse scattering problem for local spins and correlation functions

3.1 Local spin operators in the F-basis and quantum inverse problem

The spin operators in F-basis at the first or the last site of the chain can be easily obtained due to the very simple form (35) of $\tilde{F}_{1,2\dots N}$

$$F_{1\dots N} \sigma_1^- F_{1\dots N}^{-1} = \tilde{F}_{1,2\dots N} F_{2\dots N} \sigma_1^- F_{2\dots N}^{-1} \tilde{F}_{1,2\dots N}^{-1} = \tilde{D}_{1\dots N}(\xi_1; \xi_1, \xi_2, \dots, \xi_N) \sigma_1^- \quad (41)$$

and by symmetry so do the last site spin operator.

Taking into account that $\tilde{D}$ is totally symmetric, for operator on site i we obtain:

$$F_{i\dots N 1\dots i-1} \sigma_i^- F_{i\dots N 1\dots i-1}^{-1} = \tilde{D}_{1\dots N}(\xi_i) \sigma_i^- \quad (42)$$

Thus our next goal is to evaluate the product of two "permuted" F-matrices $F_{1\dots N} F_{i\dots N 1\dots i-1}^{-1}$, which can be considered as the expression in the F-basis of a propagator $F_{i\dots N 1\dots i-1}^{-1} F_{1\dots N}$.

Theorem 3.1. Let U_1^i be the propagator $F_{i\dots N1\dots i-1}^{-1}F_{1\dots N}$ from site 1 to site i of the chain. It can be written into the following forms:

$$U_1^i = R_{i-1,i\dots N1\dots i-2}\dots R_{2,3\dots N1}R_{1,2\dots N} \quad (43)$$

$$= \prod_{\alpha=1}^{i-1} (A_{1\dots N}(\xi_\alpha) + D_{1\dots N}(\xi_\alpha)) \quad (44)$$

Note that in (44), all factors commute with each other.

Remark 3.2. The propagator U_1^1 through the whole chain being the identity, note that

$$\prod_{\alpha=1}^N (A_{1\dots N}(\xi_\alpha) + D_{1\dots N}(\xi_\alpha)) = R_{N,1\dots N-1}\dots R_{1,2\dots N} = 1 \quad (45)$$

Thus the inverse $(U_1^i)^{-1}$ of the propagator on a part of the chain is merely the propagator $U_i^1 = \prod_{\alpha=i}^N (A + D)(\xi_\alpha)$ on the remaining part.

Remark 3.3. The action of propagator U_1^i on an operator entry $X_{1\dots N}$ of the monodromy matrix ($X = A, B, C, D$):

$$U_1^i X_{1\dots N} = X_{i\dots N1\dots i-1} U_1^i \quad (46)$$

or equivalently

$$U_1^i T_{0,1\dots N} = T_{0,i\dots N1\dots i-1} U_1^i \quad (47)$$

Theorem 3.1 allows us to obtain the value of any local operator in F-basis:

$$F_{1\dots N} \sigma_i^- F_{1\dots N}^{-1} = \prod_{\alpha=1}^{i-1} (\tilde{A} + \tilde{D})(\xi_\alpha) \cdot \tilde{D}(\xi_i) \sigma_i^- \cdot \prod_{\alpha=1}^{i-1} (\tilde{A} + \tilde{D})^{-1}(\xi_\alpha) \quad (48)$$

Note that $\tilde{B}$ and $\tilde{C}$ are quasilocal in terms of the σ_i^- or σ_i^+ . Indeed a direct calculus in the F-basis gives the identity:

$$\tilde{D}(\xi_i) \sigma_i^- (\tilde{A} + \tilde{D})(\xi_i) = \tilde{B}(\xi_i) \quad (49)$$

The proof of the last formula is as below.

3.2 Reconstruction of local spin operators

Theorem 3.4. Local spin operators at a given site i of the inhomogeneous XXX or XXZ Heisenberg chain are given by:

$$\sigma_i^- = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot B(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (50)$$

$$\sigma_i^+ = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot C(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (51)$$

$$\sigma_i^z = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot (A - D)(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (52)$$

Proof is by taking x_0 with $\sigma_0^-, \sigma_0^+, \sigma_0^z$ respectively for the Lemma below

Lemma 3.5. *Let x_i act on quantum spin space V_i and x_0 on auxiliary space V_0 . They are related by the identity:*

$$tr_0(x_0 R_{0,1\dots N}(\xi_i)) = \prod_{\alpha=1}^{i-1} (A+D)^{-1}(\xi_\alpha) \cdot x_i \cdot \prod_{\alpha=1}^i (A+D)(\xi_\alpha) \quad (53)$$

3.3 General formula for correlation functions

Here we give the k-point spin-spin correlation function between two Bethe states. Indeed, for any integer k and any subset $\{i_j\}_{1 \leq j \leq k}$ of $\{1 \dots N\}$, with the convention $i_1 \leq i_2 \leq i_3 \dots \leq i_k$, the correlation function for spins at sites $i_1, \dots, i_k$ between two Bethe states $\langle 0 | C(\mu_1) \dots C(\mu_{n_1})$ and $B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle$ can be written into the following form:

$$\begin{aligned} & \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \sigma_{i_1}^{\epsilon_1} \sigma_{i_2}^{\epsilon_2} \dots \sigma_{i_k}^{\epsilon_k} B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \\ &= \prod_{\alpha=1}^{i_1-1} \prod_{j=1}^{n_1} b^{-1}(\mu_j, \xi_\alpha) \cdot \prod_{\alpha=i_k+1}^N \prod_{j=1}^{n_2} b^{-1}(\lambda_j, \xi_\alpha) \\ & \times \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \cdot X^{\epsilon_1}(\xi_{i_1}) \cdot \prod_{\alpha=i_1+1}^{i_2-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_2}(\xi_{i_2}) \dots \\ & \times \dots \prod_{\alpha=i_k+1}^{i_k-1} (A+D)(\xi_\alpha) \cdot X^{\epsilon_k}(\xi_{i_k}) \cdot B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \end{aligned} \quad (54)$$

where $\epsilon_i, 1 \leq j \leq k$ takes the values $+, -, \text{ or } z$, X^{ϵ_j} being respectively to $C, B, A-D$.

4 Form factors

4.1 operators σ_m^- and σ_m^+

We begin with two one-point functions:

$$F_n^-(m, \{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^{n+1} C(\mu_j) \sigma_m^- \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (55)$$

and

$$F_n^+(m, \{\lambda_k\}, \{\mu_j\}) = \langle 0 | \prod_{k=1}^n C(\lambda_k) \sigma_m^+ \prod_{j=1}^{n+1} B(\mu_j) | 0 \rangle \quad (56)$$

where $\{\lambda_k\}_n$ and $\{\mu_j\}_{n+1}$ are solutions of the Bethe equations.

Proposition 4.1. *For two Bethe states with spectral parameters $\{\lambda_k\}_n$ and $\{\mu_j\}_{n+1}$ the matrix element of the operator σ_m^- can be represented as a determinant,*

$$\begin{aligned} F_n^-(m, \{\mu_j\}, \{\lambda_k\}) &= \frac{\phi_{m-1}(\{\mu_j\}) \prod_{j=1}^{n+1} \varphi(\mu_j - \xi_m + \eta)}{\phi_{m-1}(\{\lambda_k\}) \prod_{k=1}^n \varphi(\lambda_k - \xi_m + \eta)} \\ & \times \frac{1}{\prod_{n+1 \geq k > j \geq 1} \varphi(\mu_k - \mu_j) \prod_{1 \leq \beta \leq \alpha \leq n} \varphi(\lambda_\beta - \lambda_\alpha)} \\ & \times \det_{n+1} H^-(m, \{\mu_j\}, \{\lambda_k\}) \end{aligned} \quad (57)$$

where the coefficients $\phi_m(\{\lambda_k\})$ are

$$\phi_m(\{\lambda_k\}) = \prod_{k=1}^n \prod_{j=1}^m b^{-1}(\lambda_k, \xi_j) \quad (58)$$

and the $(n+1) \times (n+1)$ matrix H^- is defined as

$$H_{ab}^-(m) = \frac{\varphi(\eta)}{\varphi(\mu_a - \lambda_b)} \left(a(\lambda_b) \prod_{\substack{j=1 \\ j \neq a}}^{n+1} \varphi(\mu_j - \lambda_b + \eta) - d(\lambda_b) \prod_{\substack{j=1 \\ j \neq a}}^{n+1} \varphi(\mu_j - \lambda_b - \eta) \right) \quad (59)$$

for $b < n+1$

$$H_{an+1}^-(m) = \frac{\varphi(\eta)}{\varphi(\mu_a - \xi_m + \eta) \varphi(\mu_a - \xi_m)} \quad (60)$$

And $F_n^+(m, \{\lambda_k\}, \{\mu_j\})$ share a similar representation,

$$F_n^+(m, \{\lambda_k\}, \{\mu_j\}) = \frac{\phi_m(\lambda_k) \phi_{m-1}(\lambda_k)}{\phi_{m-1}(\mu_j) \phi_m(\mu_j)} F_n^-(m, \{\lambda_k\}, \{\mu_j\}) \quad (61)$$

4.2 Operator σ_m^z

We calculate

$$F_n^z(m, \{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \sigma_m^z \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (62)$$

Proposition 4.2. *The one-point function can be represented as a determinant:*

$$\begin{aligned} F_n^z(m, \{\mu_j\}, \{\lambda_k\}) &= \frac{\phi_{m-1}(\{\mu_j\})}{\phi_{m-1}(\{\lambda_k\})} \prod_{j=1}^n \frac{\varphi(\mu_j - \xi_m + \eta)}{\varphi(\lambda_j - \xi_m + \eta)} \\ &\quad \times \frac{1}{\prod_{j>k} \varphi(\mu_k - \mu_j) \prod_{\alpha<\beta} \varphi(\lambda_\beta - \lambda_\alpha)} \\ &\quad \times \det_n (H(\{\mu_j\}, \{\lambda_k\}) - 2P(m, \{\mu_j\}, \{\lambda_k\})) \end{aligned} \quad (63)$$

where

$$H_{ab} = \frac{\varphi(\eta)}{\varphi(\mu_a - \lambda_b)} \left(a(\lambda_b) \prod_{j \neq a} \varphi(\mu_j - \lambda_b + \eta) - d(\lambda_b) \prod_{j \neq a} \varphi(\mu_j - \lambda_b - \eta) \right) \quad (64)$$

$1 \leq a, b \leq n$

And $P(m, \{\mu_j\}, \{\lambda_k\})$ is a matrix of rank one,

$$P_{ab}(m) = \phi(\eta) \frac{\prod_{k=1}^n \phi(\lambda_k - \lambda_b + \eta)}{\phi(\mu_a - \xi_m) \phi(\mu_a - \xi_m + \eta)} \quad 1 \leq a, b \leq n \quad (65)$$

Proposition 4.3. *The σ_z form factor in the thermodynamic limit between the same bethe state. The form factor is defined as:*

$$\frac{\langle 0 | \prod_{j=1}^N C(\lambda_j) \sigma_m^z \prod_{j=1}^N B(\lambda_j) | 0 \rangle}{\langle 0 | \prod_{j=1}^N C(\lambda_j) \prod_{j=1}^N B(\lambda_j) | 0 \rangle} \quad (66)$$

We have by inverse scattering problem:

$$\sigma_m^z = 2 \prod_{j=1}^{m-1} (A + D)(\xi_j) A(\xi_m) \prod_{m+1}^N (A + D)(\xi_j) - I \quad (67)$$

We would like to calculate firstly the main part

$$\begin{aligned} & \langle 0 | \prod_{j=1}^n C(\lambda_j) \prod_{j=1}^{m-1} (A + D)(\xi_j) A(\xi_m) \prod_{m+1}^N (A + D)(\xi_j) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \\ &= \prod_{n=1}^n \frac{\varphi(\lambda_n - \xi_m)}{\varphi(\lambda_n - \xi_m + \eta)} \langle 0 | \prod_{j=1}^n C(\lambda_j) A(\xi_m) \prod_{j=1}^n B(\lambda_j) | 0 \rangle \end{aligned} \quad (68)$$

And we have

$$\begin{aligned} & \langle 0 | \prod_{j=1}^n C(\lambda_j) A(\xi_m) \\ &= \sum_{a'=1}^{N+1} a(\lambda_{a'}) \frac{\prod_{k=1}^N \varphi(\lambda_k - \lambda_{a'} + \eta)}{\prod_{\substack{k=1 \\ k \neq a'}}^{N+1} \varphi(\lambda_k - \lambda_{a'})} \langle 0 | \prod_{\substack{k=1 \\ k \neq a'}}^{N+1} C(\lambda_k) \end{aligned} \quad (69)$$

Before proceeding, we firstly calculate

$$\begin{aligned} & \langle 0 | \prod_{k=1}^{N-1} C(\lambda_k) C(\xi_m) \prod_{k=1}^N B(\lambda_k) | 0 \rangle \\ &= \end{aligned} \quad (70)$$